ModelIt!

Teacher's Guide

Chain Reaction: When Energy Moves Object to Object

Why Each Collision Loses a Little Energy Along the Way

Quick Reference

Grade Level	Grade 04L2
NGSS Standard	4-PS3-3, 4-PS3-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-PS3-3, 4-PS3-4

Ask questions and predict outcomes about the changes in energy that occur when objects collide. Apply scientific ideas to design, test, and refine a device that converts energy from one form to another.

Three-Dimensional Learning

Science Practice	**Developing and Using Models / Asking Questions** — Students develop a five-component model of energy transfer through collisions and ask questions about how push force and mass affect the energy available at the end of a chain.
Disciplinary Core Idea	**PS3.C Relationship Between Energy and Forces / PS3.D Energy in Chemical Processes and Everyday Life** — When objects collide, energy is transferred. The transfer is not 100% efficient — some energy becomes sound, heat, and vibration, reducing the energy available for movement.
Crosscutting Concept	**Energy and Matter** — Students trace energy flow through a chain of collisions, identifying where energy enters the system, how it transfers between objects, and where it exits as sound and heat.

CAST Testing Alignment

MULTIPLE CHOICE:

A student pushes a marble into a line of 10 identical marbles. The last marble rolls out the other end but moves slower than the first marble was pushed. Which best explains why?

CONSTRUCTED RESPONSE:

Using what you know about energy transfer and collisions, explain why a Rube Goldberg machine with 20 steps needs a stronger starting push than one with 5 steps. Use the words 'kinetic energy transfer,' 'energy loss,' and 'collision' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every time two objects collide, energy is transferred — but never perfectly. Understanding how energy moves from object to object and where it gets lost helps us explain everything from how Newton's cradles work to why car crumple zones save lives to why the last domino in a long chain falls with less force than the first.

What Happens During a Collision

When a moving object hits a stationary object, kinetic energy transfers from the moving object to the stationary one. In a perfectly elastic collision (like ideal billiard balls), all kinetic energy transfers and no energy is lost to other forms. In reality, perfectly elastic collisions do not exist. Every real collision converts some kinetic energy into sound (the bang or click you hear), heat (friction and deformation at the contact point), and vibration (the objects shake briefly after impact).

Energy Loss at Every Step

In a chain reaction with multiple collisions, energy is lost at EACH collision point. If 5% of kinetic energy is lost as sound and heat at each collision, then after 1 collision you have 95% of the original energy, after 2 collisions about 90%, after 5 collisions about 77%, and after 10 collisions only about 60%. This is why long chain reactions require a strong initial push — there must be enough energy at the start to survive all the losses and still have enough at the end to complete the final action.

Mass Matters

The mass of the objects in a chain dramatically affects energy transfer. When a moving object collides with a stationary object of the SAME mass, nearly all kinetic energy transfers (minus losses). When a light object hits a heavy one, most energy bounces back and little transfers to the heavy object. When a heavy object hits a light one, the heavy object barely slows down and the light one shoots away with lots of speed. This is why Newton's cradles use identical balls — same mass ensures maximum energy transfer.

Real-World Chain Reactions

Chain reactions are everywhere in the real world. In bowling, the ball transfers energy to the first pin, which transfers energy to pins behind it in a chain reaction. In car accidents, crumple zones are designed to absorb collision energy through a chain of controlled deformations, protecting the people inside. In nuclear reactions, one atom splitting releases energy that triggers neighboring atoms to split — a chain reaction that powers nuclear plants. Even your body uses chain reactions — nerve signals pass from cell to cell in a chain.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Initial Push Force and Object Mass as external components (controllable choices) and Kinetic Energy Transfer, Energy Lost to Sound, and Final Object Speed as internal components that result from the collision chain.

E **Establish Relationships**

Students determine that Initial Push Force positively affects Kinetic Energy Transfer, Object Mass negatively affects Final Object Speed (heavier objects are harder to move), KE Transfer positively affects both Final Object Speed and Energy Lost to Sound, and Energy Lost to Sound negatively affects Final Object Speed.

V **Visualize & Model**

Students build a five-component model showing how energy flows from an initial push through a chain of collisions, with each collision leaking energy as sound, until what remains determines the final object's speed.

E **Evaluate Outcomes**

Students run four scenarios to observe how push force and object mass combine to determine how much energy survives the chain, discovering that both strong pushes and light objects maximize final speed.

R **Revise & Extend**

Students add number of objects in chain or object material to explore how chain length amplifies energy loss and how different materials transfer energy with different efficiency.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Watch this — I am going to push ONE marble, and something is going to happen at the OTHER end of this line. How does the energy get from here to there?"*

Watch For

Demonstrate a Newton's cradle or a line of marbles. Pull back one ball, let it go, and watch the last ball fly out. Ask: How did the energy travel through all those balls?

Slide 2: Learning Objectives

3 min

Say: *"Today we are tracking energy as it JUMPS from one object to the next — and finding out where it leaks away at every step."*

Watch For

Have students read objectives aloud. Pre-teach 'collision' and 'energy transfer' by rolling a marble into a stationary marble.

Slide 3: Big Question

3 min

Say: *"If I push the first domino hard, the last domino should fall just as hard, right? But it does not. Where does the energy GO?"*

Watch For

Set up a line of 10 dominoes. Push the first one. Ask students to watch carefully — does the last one fall with the same force? Why not?

Slide 4: LEVER Framework

2 min

Say: *"We are modeling what happens at every single collision in a chain reaction. Five components, and a sneaky energy leak."*

Watch For

Preview all five components. Tell students to listen for the 'click' sound at each collision — that click is ENERGY ESCAPING as sound.

Slide 5: Activity 1: Components

8 min

Say: *"We control two things: how hard we push and how heavy the objects are. Everything else is physics."*

Watch For

Guide sorting. Initial Push Force and Object Mass are external (we choose them). KE Transfer, Sound Loss, and Final Speed are internal (physics determines them).

Slide 6: Activity 2: Connections

10 min

Say: *"Here is the tricky part: Kinetic Energy Transfer has TWO positive arrows going out — one to Final Object Speed and one to Energy Lost to Sound. They are COMPETING for the same energy!"*

Watch For

Help students map the relationships. Emphasize that KE Transfer feeds both final speed AND sound loss — more energy transferring also means more energy leaking.

Slide 7: Activity 3: Simulation

12 min

Say: *"Watch the energy graph as the chain runs. See how each collision takes a little bite out of the energy? By the end, there is much less."*

Watch For

Run all four scenarios. Focus on the strong/light vs. weak/heavy comparison — the difference in final speed is dramatic.

Slide 8: Discoveries

5 min

Say: *"Every collision is an energy leak. The sound you hear is PROOF that energy is escaping from the motion system."*

Watch For

Have students clap their hands — the sound they hear IS kinetic energy transforming into sound energy. Each collision does the same thing.

Slide 9: STEM Challenge

12 min

Say: *"Build the LONGEST chain reaction you can where energy transfers through at least five collisions and completes a final action at the end!"*

Watch For

Distribute materials (marbles, dominoes, toy cars, blocks, ramps). Groups design, build, and test chain reactions. Count successful collisions.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Initial Push Force, Object Mass
- **INTERNAL:** Kinetic Energy Transfer, Energy Lost to Sound, Final Object Speed

Relationships

- **Initial Push Force to Kinetic Energy Transfer:** POSITIVE (+) — A harder push gives the first object more kinetic energy, which means more energy is available to transfer through the chain. Double the push force and you roughly double the kinetic energy entering the system.
- **Object Mass to Final Object Speed:** NEGATIVE (-) — Heavier objects require more energy to accelerate. In a chain of heavy objects, more energy is absorbed by each object's inertia, leaving less energy available for the final object's speed. Light objects transfer energy more efficiently.
- **Kinetic Energy Transfer to Final Object Speed:** POSITIVE (+) — More kinetic energy transferring through the chain means more energy reaches the final object. The final object's speed depends directly on how much energy survives all the collisions to reach it.
- **Kinetic Energy Transfer to Energy Lost to Sound:** POSITIVE (+) — When more energy transfers during a collision, the collision is also more energetic — producing a louder sound. More energy flowing through the system means more energy leaking out as sound at each collision point.
- **Energy Lost to Sound to Final Object Speed:** NEGATIVE (-) — Energy converted to sound is energy that is no longer available for motion. Every bit of energy that becomes sound at each collision is subtracted from the energy that could have reached the final object. More sound loss means slower final object.

Key Discoveries

- Energy transfers from object to object during collisions, but some energy is LOST to sound and heat at each collision point
- Heavier objects require more energy to get moving, so less kinetic energy transfers through a chain of heavy objects compared to light ones
- A stronger initial push gives the chain more total energy to work with, so more energy survives the losses at each collision
- Every chain reaction gets weaker over time because energy is lost at every collision — this is why the last domino falls with less force than the first one was pushed

Multiple Choice Post-Assessment Answer Key

1. When a moving marble hits a row of stationary marbles, the last marble in the row rolls away. What happened to the energy?

Correct: B) The kinetic energy from the first marble transferred through the chain from marble to marble until it reached the last one

Correct! Kinetic energy (motion energy) transfers from one object to the next during each collision. The first marble's energy passes through the chain until the last marble receives it and moves.

2. In the model, Initial Push Force and Object Mass are external components. Why are these external?

Correct: B) Because they are the starting conditions we control, and the internal components respond to them

Right! External components are the inputs we set before the system runs. We decide how hard to push the first object and how heavy the objects are. Kinetic Energy Transfer, Energy Lost to Sound, and Final Object Speed all respond to these choices.

3. Every time one domino hits the next in a chain, you hear a clicking sound. What does that clicking sound tell you about energy?

Correct: B) The sound is proof that some kinetic energy is transforming into sound energy at each collision, reducing the energy available for motion

Exactly! Every click you hear represents kinetic energy that transformed into sound energy. That energy is no longer available to push the next domino. This is why chain reactions lose energy at every collision.

4. The model shows that Object Mass has a NEGATIVE relationship with Final Object Speed. A student tests the chain with light marbles and then with heavy steel balls. What will they observe?

Correct: C) The light marble chain will have a faster Final Object Speed because lighter objects transfer energy more efficiently and require less energy to get moving

Right! Lighter objects require less energy to accelerate, so more kinetic energy passes through the chain. Heavier objects absorb more energy just getting started, leaving less energy for the final object. The negative relationship means more mass leads to less final speed.

5. The model shows that Energy Lost to Sound has a NEGATIVE relationship with Final Object Speed. Why does a long chain of 20 dominoes produce a slower final domino than a short chain of 5 dominoes with the same starting push?

Correct: B) A longer chain has more collisions, and each collision loses energy to sound and heat, so less energy reaches the final domino

That is right! Every collision is an energy leak. With 20 collisions, the energy losses add up. Each click of sound and tiny bit of heat removes energy from the motion system. By the time energy reaches domino 20, much less is left than what domino 5 received.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Energy passes through the chain without any loss"	Every real collision loses some energy to sound, heat, and vibration. You can HEAR this loss — the clicking or banging sound during each collision IS kinetic energy that has transformed into sound energy and left the motion system. If energy transferred perfectly, every chain reaction would go on forever, but they do not because of these unavoidable losses.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Heavier objects are better for chain reactions because they have more energy"	While heavier objects CAN carry more kinetic energy when moving at the same speed, they also require MORE energy to get moving in the first place. In a chain reaction, a heavy stationary object absorbs more of the incoming energy just to start moving, leaving less to pass to the next object. LIGHTER objects require less energy to accelerate, so more energy flows through the chain.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The sound of a collision is separate from the energy transfer"	The sound IS part of the energy transfer — or more accurately, it is energy LEAKING from the transfer. When two objects collide, kinetic energy transforms into several forms simultaneously: some transfers as kinetic energy to the next object, some becomes sound waves, and some becomes heat. The sound and heat are not separate events — they are part of the same collision physics. Louder collisions mean more energy was converted to sound, meaning LESS energy was available for motion.